

Human-Aligned Sidewalk Accessibility Perception: Soft-Label Learning from 829 Mobility Aid Users for Autonomous Mobility Services

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Abstract: Before a shared autonomous vehicle opens its door at a curbside drop-off location, the planner must answer the question if the adjacent sidewalk is accessible for the passenger’s specific mobility aid. Rather than collapsing the problem into binary labels, we instead frame the issue as calibrated perception for risk-aware pedestrian routing. Using 49,490 ternary responses (yes/unsure/no) from 829 mobility-aid users on 52 Google Street View scenes via Project Sidewalk, it yields an empirical per-aid vote distributions that we use as the operational training target. A frozen vision encoder with a per-aid linear probe trained by KL divergence against these distributions achieves a soft Brier score of 0.076 (mean across eight backbones). This results in a $3.2\times$ calibration improvement over standard cross-entropy that holds for every encoder tested. To demonstrate this, we integrate these calibrated probabilities into an OpenStreetMap graph to power a risk-aware routing algorithm. The planner effectively trades a minor distance overhead (7.4–9.8%) for safer drop-off locations, improving overall passability on 87.8–91.0% of trips. In a 50-pair head-to-head comparison, our approach outperforms OpenRouteService’s (ORS) standard wheelchair profile across all mobility aids, and successfully maps the 20% of routes where ORS fails to find any path at all. We frame this module as a learned map-prior layer for AV drop-off planning, seamlessly integrating with onboard camera/LiDAR verification. Additionally, it operates two orders of magnitude faster than the closest zero-shot VLM, achieving a latency of just 78.5 ms/image on an A100 GPU for the strongest backbone.

Keywords: Accessibility Perception, Soft-Label Learning, Risk-Aware Routing

1 Introduction

A shared autonomous vehicle preparing for a curbside drop-off must answer a question the onboard map almost never resolves: *is the sidewalk segment between the door and the passenger’s destination passable for that passenger’s specific mobility aid?* This last-meter perception problem [1, 2, 3] sits between two well-studied layers of the autonomy stack: the geometric perception running onboard [4, 5, 6] and the rule-based pedestrian-routing engine that consumes city-scale maps [7, 8]. The issue is that neither of these existing systems is calibrated for the actual risk experienced by a mobility-aid user. What one manual-wheelchair user judges traversable, another may judge impassable for entirely correct reasons [9]. If a navigation planner receives this data as a continuous probability rather than a simple binary “yes/no” label, it can assess whether it should trade a shorter, riskier path for a longer, safer one. Therefore, the core robot-learning question is not “can a network classify sidewalks,” but *can a learned perception module generate a calibrated, per-aid probability that an autonomous planner can reliably use to navigate uncertainty?*

Perception under irreducible human disagreement. Two experienced wheelchair users looking at the same image of a cracked, sloped sidewalk may reach opposite conclusions (Fig. 1a). This

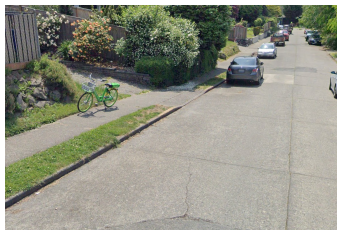
is not annotator noise, as it reflects real variation in physical ability, equipment, and acceptable risk [9, 10]. Treating it as noise, as nearly every prior automated-accessibility work has done [11, 5, 12, 13] via majority-vote collapsing, discards exactly the signal that a risk-aware planner needs. Modern networks are also known to be systematically overconfident on subjective inputs [14, 15, 16]. However, a lesson from Peterson et al. [17] on the CIFAR-10H dataset shows how models become more reliable when trained on distributions of human uncertainty rather than rigid binary labels.

Our position. We treat the empirical vote distribution $[p_{\text{no}}, p_{\text{unsure}}, p_{\text{yes}}]$ over {no, unsure, yes} as the operational reference signal for risk-aware autonomous mobility services (Fig. 1b). With ≈ 190 independent votes per (image, aid) pair, nearly four times the per-image annotation density of CIFAR-10H [17] and far richer than ImageNet-16H [18], each distribution is a statistically reliable empirical estimate of community-level opinion, not a noisy majority vote from three crowdworkers. A frozen vision encoder + per-aid linear probe trained against these distributions by KL divergence produces a calibrated per-aid probability that we drop directly into an OpenStreetMap (OSM) derived Pittsburgh pedestrian graph as edge cost. The planner solves a single risk-aware Dijkstra and produces five *different* optimal paths for the same origin-destination (OD) pair, one per aid (Fig. 1c). The metric that matters for this pipeline is calibration (soft Brier score), *not* argmax F1: the planner’s edge weight is a linear function of $\hat{p}_{\text{yes}}$, so an overconfident probe collapses the routing gradient even when argmax accuracy is high.

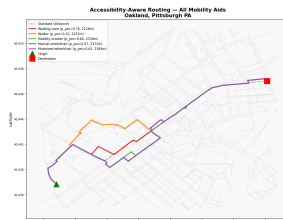
Contributions. (C1) We develop a learned perception module that trains a frozen-encoder per-aid linear probe against 829-user vote distributions via KL divergence. This approach yields a $3.2\times$ soft-Brier calibration improvement over hard cross-entropy—a win that remains perfectly consistent across eight different encoders. (C2) We expose exactly why hard labels fail for accessibility perception under class imbalance. Minority images (like those for walking canes) often trigger tied human votes; forcing these into binary labels causes hard cross-entropy to drop 6/8 encoders below a trivial baseline. Our Soft-KL approach recovers +42% relative F1, revealing a structural lesson that applies to any robot-learning task trained on subjective human judgment. (C3) We build a planner interface that treats these learned probabilities as risk-aware Dijkstra edge costs, demonstrated on 10,000 random OD pairs in Pittsburgh. The policy finds safer routes (87.8–91.0% improved passability) at a small cost to distance (7.4–9.8% overhead). It universally beats OpenRouteService’s wheelchair profile across all aids, and effortlessly routes the 20% of paths that ORS cannot route. (C4) We conduct an encoder-transfer probe in Detroit, Zurich, and Taipei, which successfully identifies a Western-centric data limitation (accuracy in Taipei drops to near chance). We use this to design a vital safeguard: ensuring a deployed robot rejects the map-prior in geographic regions where it knows it cannot be trusted



(a) **Divergent opinions.** Bumpy stone pavers near a curb cut produce a manual-wheelchair split of 47% no, 9% unsure, and 44% yes, while walking-cane users vote 76% yes.



(b) **Soft-label perception.** A frozen encoder plus linear probe predicts per-aid distributions $[\hat{p}_{\text{no}}, \hat{p}_{\text{unsure}}, \hat{p}_{\text{yes}}]$ instead of binary classes.



(c) **Routing.** Calibrated per-aid edge probabilities in a Pittsburgh graph yield distinct routes per aid and beat OpenRouteService on all five aids.

Figure 1: From vote distributions to accessibility-aware routing. Crowdsourced disagreement is signal, not noise. We train a calibrated per-aid probe from soft labels, embed the resulting probabilities into a pedestrian graph, and demonstrate a routing system that produces a different optimal path for each mobility aid.

73 2 Related Work

74 **Automated accessibility perception and pedestrian routing.** A decade of work crowdsources
 75 sidewalk audits from Street View [19, 20, 11, 4], scales them with deep classifiers [5, 12], and com-
 76 plements them with aerial sidewalk extraction [6, 13] and multimodal robot/phone sensing [21, 22].
 77 A separate line builds map-level routing engines that consume accessibility metadata: OpenRoute-
 78 Service’s wheelchair profile [7] parses OSM tags into edge costs, AccessMap and the OpenSide-
 79 walks schema [1, 8] formalise per-user pedestrian network data, MyPath [2] adds survey-elicited
 80 per-user weights, and Tannert and Schöning [3] document the tens-of-percent distance penalty
 81 wheelchair users pay today. Every prior accessibility classifier trains on hard labels, and every
 82 routing engine relies on sparse rule-based OSM tags. We learn a calibrated per-aid probability from
 83 *mobility-aid users’ own* vote distributions and use it as a drop-in Dijkstra edge cost, covering 91.7%
 84 of edges via direct image inference where ORS-style tag lookup leaves 20% of OD pairs unroutable.

85 **Foundation encoders for robot perception.** CLIP [23], DINOv2 [24], and SigLIP2 [25] have be-
 86 come standard backbones for transfer in robot-learning systems where downstream data is scarce.
 87 We evaluate all three families plus supervised ViT [26] on the same folds; the calibration gap be-
 88 tween training objectives we report is essentially backbone-invariant.

89 **Soft labels and calibration for safety-critical learning.** Label Distribution Learning [27] and the
 90 CIFAR-10H benchmark [17] established empirical vote distributions as first-class training targets,
 91 with subsequent extensions to ImageNet-16H [18], sparse-annotator regimes [10], crowd-distance
 92 selective prediction [16], and broad task coverage [28]. Soft-label distillation [29] is structurally
 93 identical but uses a teacher model rather than humans as the source. In parallel, Guo et al. [14] and
 94 Minderer et al. [15] showed that modern networks are systematically overconfident on subjective
 95 inputs, and Zollo et al. [30] make calibration the primary safety target for embodied agents whose
 96 downstream planner consumes probability outputs. Our contribution sits at that intersection: the soft
 97 Brier metric is selected by the planner, not by aesthetic preference.

98 **Zero-shot VLMs as a perception alternative.** Instruction-tuned VLMs [31, 32, 33, 34] are increas-
 99 ingly proposed as drop-in perception for robotic tasks. We provide the first latency- and calibration-
 100 grounded comparison against trained encoder probes on the accessibility task: even the strongest
 101 VLM trails our best probe in calibration by $6\times$ and in latency by $\sim 80\times$.

102 3 Dataset and Soft-Label Formulation

103 **Data.** Project Sidewalk [4, 9] supplies 49,490 ternary responses (yes/unsure/no) from 829 mobility-
 104 aid users on 52 rectilinear 640×640 Google Street View deprojections, $4.4\times$ the CHI 2025 release
 105 on the same platform. Each view is rated under each of five aids (manual wheelchair, motorized
 106 wheelchair, mobility scooter, walker, walking cane), giving 260 (image, aid) pairs with ≈ 190
 107 votes/pair (range 141–240). The 52 views span ten cities (Amsterdam, CDMX, Chicago, Columbus
 108 OH, La Piedad MX, Oradell NJ, Seattle, St. Louis, Teaneck NJ, Pittsburgh); the same pixel-identical
 109 images are shown to humans and to the model. A YOLOv12-seg [35] area-ratio check $A_{\text{pred}}/A_{\text{gt}} \in$
 110 $[0.7, 1.3]$ filters ~ 60 candidates to the final 52; train/test splits are panorama-level at seed 42.

111 **Vote aggregation and soft labels.** For each pair (i, a) with K votes we form $\mathbf{y}_{i,a} =$
 112 $K^{-1} \sum_k \text{one_hot}(v_{k,a}) \in \Delta^2$, an empirical distribution over {no, unsure, yes}. Its entropy
 113 $H(\mathbf{y}_{i,a}) = -\sum_c y_{i,a}^{(c)} \log y_{i,a}^{(c)}$ defines the sample weight $w_{i,a} = 1 - H(\mathbf{y}_{i,a})/H_{\text{max}}$ that down-
 114 weights highly ambiguous examples during training. Vote entropy is high across all five aids (mean
 115 normalised entropy 0.77–0.90; Appendix A, Fig. A.1), confirming that accessibility is genuinely
 116 probabilistic rather than mislabeled.

117 **Why a small visual set is appropriate.** The probe (§4) has only $3(d+1)$ parameters per aid ($d =$
 118 $1024 \Rightarrow 3075$ for DINOv2-large) and is fit against 52 vector-valued targets per aid; each target
 119 component is the mean of ~ 60 – 70 Bernoulli draws (binomial 95% CI ± 0.06 – 0.13). Pixel-level
 120 features come from foundation encoders pretrained on 10^8 – 10^9 images [23, 24, 25]; the probe learns

an aid-conditioned linear projection onto the 3-simplex. The calibration claim is therefore local to the support of these 52 scenes—we treat the visual sample size as the headline limitation in §8. Walking cane shows the strongest yes-skew (45/52 majority-yes, 6.4:1) and motorised wheelchair is evenly split (26/26); the imbalance asymmetry drives the per-aid behaviour analysed in §5.

4 Method

4.1 Linear Probe on Frozen Encoders

Each image x_i is embedded by a frozen encoder E and L2-normalised: $\mathbf{f}_i = E(x_i)/\|E(x_i)\|_2$. A per-aid linear head produces predictions:

$$\hat{\mathbf{y}}_{i,a} = \text{softmax}(W_a \mathbf{f}_i + b_a). \quad (1)$$

Encoder weights are never updated. We evaluate eight encoders: CLIP ViT-B/32, ViT-B/16, and ViT-L/14 [23]; DINOv2-base and DINOv2-large [24]; SigLIP2-base and SigLIP2-SO400M [25]; and supervised ViT-B/16 from timm [26].

4.2 Soft-KL Divergence Training

The primary loss minimises the entropy-weighted KL divergence from model output to human vote distribution:

$$\mathcal{L}_{\text{Soft-KL}} = \sum_{i,a} w_{i,a} D_{\text{KL}}(\mathbf{y}_{i,a} \parallel \hat{\mathbf{y}}_{i,a}). \quad (2)$$

Unlike knowledge distillation, where soft targets come from a teacher model, the targets here are empirical human distributions backed by ≈ 190 independent judgments each. The predicted $\hat{\mathbf{y}}_{i,a}$ is directly interpretable as “the fraction of users in our study who would consider this sidewalk accessible.”

4.3 Hard Cross-Entropy Baseline

For comparison, we collapse each vote distribution to its argmax label $\hat{c}_{i,a} = \arg \max \mathbf{y}_{i,a}$ and train with standard cross-entropy:

$$\mathcal{L}_{\text{Hard-CE}} = \sum_{i,a} w_{i,a} \text{CE}(\hat{c}_{i,a}, \hat{\mathbf{y}}_{i,a}). \quad (3)$$

The entropy weight is retained so the comparison isolates the effect of the target (distribution vs. hard label) rather than the weighting.

4.4 Calibration Metric: Soft Brier Score

We assess calibration against the human vote distribution:

$$B_{\text{soft}} = \frac{1}{N} \sum_{i,a} \|\hat{\mathbf{y}}_{i,a} - \mathbf{y}_{i,a}\|_2^2. \quad (4)$$

Lower is better. A model with $B_{\text{soft}} = 0.07$ that outputs $\hat{p}_{\text{yes}} = 0.65$ for a sidewalk where 62% of users voted “yes” is providing actionable probabilistic information. A model with $B_{\text{soft}} = 0.23$ outputting $\hat{p}_{\text{yes}} = 0.97$ for the same image misleads any downstream planner that treats the score as a probability.

4.5 Training Details

All probes use AdamW (lr = 10^{-3} , weight decay = 10^{-4} , 50 epochs, batch size 64). Encoder weights are frozen. **5-fold cross-validation** at the panorama level prevents leakage between rectified crops from the same location. Global seed = 42. All experiments run on a single NVIDIA A100 40 GB GPU. Figure 2 summarizes the full pipeline; pseudocode for the training loop and the compute/reproducibility details are in Appendix H (Algorithm 1)

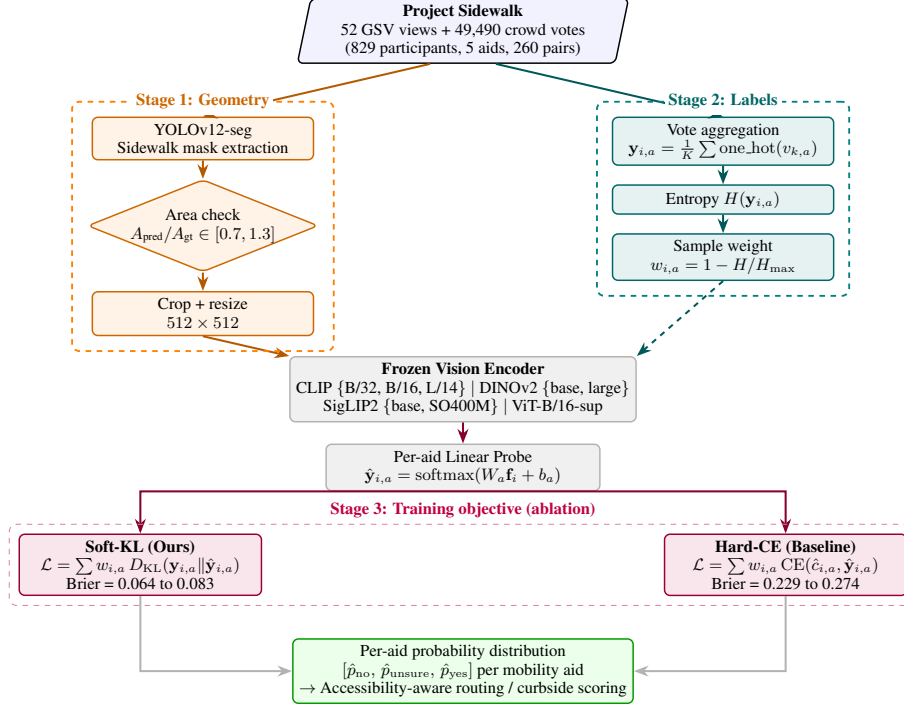


Figure 2: End-to-end pipeline. Geometry verification (orange) and entropy-weighted soft labels (teal) feed a frozen vision encoder and per-aid probe. Soft-KL preserves calibrated per-aid distributions for routing, while Hard-CE is the collapsed-label baseline.

5 Results: Calibrated Learned Perception

All numbers below are 5-fold panorama-level CV at seed 42; $\pm$ values are 95% CIs over the 40 (fold, aid) measurements per encoder. The claim is not visual foundation-model learning but whether frozen urban-scene features can be calibrated against dense human vote distributions to produce a probability a planner can consume.

Calibration headline. Table 1 and Fig. B.2 (Appendix B) report the primary calibration result alongside argmax F1 for the same encoders and folds. Soft-KL achieves a soft Brier score of 0.064–0.083 versus 0.228–0.274 for Hard-CE—a $3.2\times$ **mean improvement** that is consistent across every backbone and outside the 95% CI for all eight pairings. The Hard-CE family is overconfident in the manner Guo et al. [14] document for modern networks: predictions concentrate near $\hat{p} \in \{0, 1\}$ regardless of vote density. Operationally, at our chosen $\beta_c=8$ (§6), an edge with $\hat{p}_{\text{yes}}=0.55$ incurs $4\times$ the accessibility penalty of one with $\hat{p}_{\text{yes}}=0.91$, so the Hard-CE overconfidence collapses the planner’s gradient; Soft-KL preserves the empirical bimodality and gives the planner a usable continuous risk signal.

Why F1 understates the win. Hard-CE yields higher argmax F1 in six of eight encoders (+0.059 on average) but the gap lies inside the 95% CI in every case. The asymmetry is structural: no image in the dataset has “unsure” as the majority vote, so any probability mass Soft-KL faithfully places there is unrecoverable under argmax F1. The encoder ranking also *inverts* between objectives—DINOv2-large leads Soft-KL while CLIP-B/32 leads Hard-CE—indicating that DINOv2’s dense self-supervised features better capture the uncertainty structure of human accessibility judgments while CLIP’s contrastive language-image features better predict the plurality label.

Walking cane: a structural Hard-CE failure preserved by Soft-KL. Walking cane has a 6.4:1 positive skew (45/52 majority-yes). Under Hard-CE, every encoder scores at or below the trivial majority-class baseline of 0.467, with the best result (0.513) only marginally above it; 6/8 encoders fall *below* the baseline. Soft-KL on DINOv2-large recovers $F1 = 0.640$ (+42% relative). The mechanism: 5/7 minority-“no” walking-cane images have human vote margins < 0.05 between the

Table 1: Per-encoder results, Soft-KL (ours) vs. Hard-CE baseline. 5-fold panorama-level CV at seed 42; $\pm$ is 95% CI over 40 (fold, aid) measurements. F1 = macro-F1; Brier $\downarrow$ is the soft Brier score against human vote distributions. The bottom block reports zero-shot VLMs evaluated without any task-specific data. The Brier gap is large and consistent across all eight backbones; F1 differences are within their CIs.

Method	F1 Soft-KL	F1 Hard-CE	Brier Soft-KL	Brier Hard-CE	Ratio	Lat. (ms)
CLIP B/32	0.530 ± 0.067	0.617 ± 0.059	0.064 ± 0.011	0.231 ± 0.016	$3.6\times$	61.6
CLIP B/16	0.499 ± 0.060	0.544 ± 0.068	0.080 ± 0.017	0.239 ± 0.025	$3.0\times$	65.0
CLIP L/14	0.531 ± 0.065	0.595 ± 0.064	0.080 ± 0.011	0.230 ± 0.022	$2.9\times$	92.4
DINOv2-base	0.496 ± 0.064	0.614 ± 0.069	0.080 ± 0.010	0.238 ± 0.024	$3.0\times$	64.8
DINOv2-large	0.613 ± 0.071	0.613 ± 0.059	0.075 ± 0.013	0.253 ± 0.023	$3.4\times$	78.5
SigLIP2-base	0.502 ± 0.057	0.596 ± 0.060	0.082 ± 0.011	0.228 ± 0.020	$2.8\times$	72.1
SigLIP2-SO400M	0.579 ± 0.064	0.555 ± 0.064	0.066 ± 0.014	0.256 ± 0.022	$3.9\times$	110.0
ViT-B/16-sup	0.482 ± 0.073	0.573 ± 0.070	0.083 ± 0.015	0.274 ± 0.027	$3.3\times$	60.0
<i>Probe mean</i>	0.529	0.588	0.076	0.244	$3.2\times$	–
<i>Zero-shot VLMs (no task-specific data; F1 / Brier are single-pass)</i>						
Qwen3-VL-8B	–	0.576	–	0.468	–	6,182
LLaVA-1.5-7B	–	0.418	–	0.423	–	135
Qwen2.5-VL-7B	–	0.412	–	0.632	–	185

top two classes (Appendix D, Fig. D.1)—near-50/50 splits that Hard-CE coerces into hard negatives, creating contradictory training signal that collapses the model to the majority class. Soft-KL assigns those images targets $\approx [0.45, 0.05, 0.45]$ and preserves the training signal from the two unambiguous “no” examples. The robotic-deployment implication is direct: Hard-CE treats near-impassable edges as confidently safe for cane users; Soft-KL flags them as uncertain so the planner routes around them.

Zero-shot VLMs trail in both calibration and latency. The strongest zero-shot VLM (Qwen3-VL-8B) reaches F1 = 0.576, within 0.041 of the best trained probe, but its Brier of 0.468 is $6.2\times$ worse than DINOv2-large + Soft-KL (0.075) and its 6.2 s/image latency is $\sim 80\times$ slower than the strongest probe (and $\sim 100\times$ the lightest). LLaVA-1.5 and Qwen2.5-VL collapse to majority-class bias (balanced accuracy near 0.5). A learned per-aid probe, not a zero-shot prompt, is the configuration a robot planner can consume. Per-prompt sensitivity and the latency Pareto frontier are reported in Appendix C.

Geographic transfer probe. As an auxiliary encoder-transfer check, distinct from the per-aid task, we evaluate DINOv2-large Soft-KL on binary CurbRamp labels in three unseen cities (Appendix E, Fig. E.1). Balanced accuracy: Pittsburgh 0.566 ($n = 162$), Detroit 0.570 ($n = 91$), Zurich 0.616 ($n = 89$), Taipei 0.518 ($n = 95$). Pittsburgh/Detroit consistency rules out city-specific overfitting; Zurich gains +11.6pp likely because Amsterdam (similar infrastructure) is in the training distribution; Taipei is only +1.8pp above chance, a Western-centric-data limit we surface as the constraint on safe deployment (§8).

6 Robot-Planner Interface: Risk-Aware Routing

The learned per-aid probability $\hat{p}_{\text{yes}}^a$ is the perception module’s contract with the autonomous-mobility planner. We compose it as an OSM edge cost and show, on a real Pittsburgh pedestrian graph, that risk-aware Dijkstra yields meaningfully different routes per mobility aid and outperforms a state-of-the-art rule-based wheelchair routing engine. This is an offline planner-facing evaluation: a deployed AV would still verify the immediate curbside scene with onboard sensors before opening the door (Fig. 1c).

Graph and edge scoring. A fixed Pittsburgh OSM graph (2,046 nodes, 2,618 undirected edges, single GraphML snapshot) gives the planner topology. We assign per-aid edge priors in three stages: 40.4% of edges match Project Sidewalk GPS labels directly; 51.3% are scored by querying the nearest labeled Street View panorama and running DINOv2-large Soft-KL inference; the remaining 8.3% fall back to population-level priors. Total label-or-image coverage is 91.7%. The inferred scores are not constants—mean $\hat{p}_{\text{yes}}$ across edges ranges from 0.479 (manual wheelchair) to 0.668

(walking cane), with 49.5% of manual-wheelchair edges below 0.5. This spread is precisely what creates a usable routing gradient. The cross-city transfer probe (Appendix E) is consistent with Pittsburgh GSV being in-distribution against the audit views while Taipei sits outside, matching the transfer-probe limit above.

Routing objective and miscalibration cost. For aid a , edge e has cost $\ell_e \cdot (1 + \beta_c (1 - \hat{p}_{\text{yes},e}^a))$ (pseudocode in Appendix H, Algorithm 2); we use $\beta_c=8$ at the inflection of the Pareto sweep (Appendix F; $\beta_c \in \{1, 2, 4, 8, 16\}$ gives mean $\Delta \bar{p}_{\text{yes}}$ rising from 0.058 to 0.105 at 2.0–11.3% distance overhead). Replacing Soft-KL scores with Hard-CE scores at the same β_c produces structurally different (and uniformly worse on the calibrated metric) routes for all five aids; Hard-CE *overestimates* its own route’s $\hat{p}_{\text{yes}}$ by +0.055 to +0.085 because of overconfident edge probabilities. A robot planner consuming Hard-CE outputs would therefore prefer routes that are actually lower-quality than Soft-KL alternatives (full breakdown in Appendix G, Table G.2).

Per-aid routes (canonical OD pair). Table 2 and Fig. 1c show the five per-aid paths for one OD pair in the Forbes/Murray corridor: each aid traces a structurally distinct optimal route (+0.134 to +0.155 mean $\hat{p}_{\text{yes}}$ at 18–25% extra distance). This per-aid diversity is only possible when the planner consumes a real per-edge per-aid probability, not a single binary tag.

Table 2: Canonical OD pair ($\beta_c=8$). Standard-route baselines: 0.442–0.516.

Aid	$\bar{p}_{\text{yes}}^{\text{acc}}$	Δp_{yes}	$\Delta \ell$
Cane	0.671	+0.155	+25%
Walker	0.630	+0.144	+25%
Scooter	0.600	+0.141	+18%
Man. WC	0.576	+0.134	+19%
Mot. WC	0.592	+0.146	+25%

vs. OpenRouteService wheelchair profile. ORS [7] is the natural rule-based counterpart—it parses OSM wheelchair tags into edge costs and is the engine downstream AccessMap-style products [1, 8, 2] would compose with. On 50 random OD pairs in our graph, ORS fails to return a route for 20% of pairs (insufficient OSM tags; a known coverage failure [3]). On the 40 routable pairs, our learned per-aid scoring yields a higher mean $\hat{p}_{\text{yes}}$ on every aid (Table 3; full Std/ORS/Ours breakdown in Appendix G, Table G.1); paired-difference 95% CIs are positive for all five aids though wide at $n=40$, so we read this as directional evidence at one snapshot. The mechanism is coverage: our scoring populates 91.7% of edges with calibrated probabilities, whereas ORS depends on tags absent on most edges of most cities.

Table 3: vs. ORS on the 40/50 routable Pittsburgh pairs. Ours yields higher mean $\hat{p}_{\text{yes}}$ on every aid.

Aid	$\bar{p}_{\text{yes}}^{\text{ORS}}$	$\bar{p}_{\text{yes}}^{\text{ours}}$	Δ (95% CI)
Cane	0.712	0.748	+0.036 $\pm$ 0.018
Walker	0.634	0.668	+0.034 $\pm$ 0.020
Scooter	0.585	0.625	+0.040 $\pm$ 0.019
Man. WC	0.519	0.566	+0.047 $\pm$ 0.022
Mot. WC	0.545	0.590	+0.045 $\pm$ 0.022

Monte Carlo over 10,000 OD pairs. To verify generality, we sample 10,000 random OD pairs (min separation 200 m, seed 42) and compare standard vs. risk-aware Dijkstra (Table 4; the full P25/P50/P75 distribution and per-aid Pareto scatter are reported in Appendix F). For every aid, 87.8–91.0% of trips improve in mean $\hat{p}_{\text{yes}}$ at 7.4–9.8% mean distance overhead; the median overhead is 4.8–6.8%, indicating most trips reroute at modest cost. This is the behaviour an autonomous mobility planner needs: a hedge that is rarely large in distance but consistently positive in expected passability.

Table 4: Monte Carlo over 10,000 random OD pairs ($\beta_c=8$, seed 42, min 200 m). $\pm$ are 95% Wilson CIs.

Aid	$\Delta \bar{p}_{\text{yes}}$	Med.	% impr.	$\Delta \ell$
Cane	+0.065	+0.055	87.8 $\pm$ 0.6	+7.4%
Walker	+0.081	+0.074	88.2 $\pm$ 0.6	+8.2%
Scooter	+0.102	+0.096	91.0 $\pm$ 0.6	+8.6%
Man. WC	+0.120	+0.112	90.9 $\pm$ 0.6	+9.8%
Mot. WC	+0.116	+0.110	90.8 $\pm$ 0.6	+9.8%

7 Discussion

Calibration is the metric the robot planner selects. The Dijkstra objective is linear in $\hat{p}_{\text{yes}}$, so Hard-CE’s overconfidence [14, 15] directly distorts routing costs. A borderline edge with 62% yes-votes receives $\hat{p}_{\text{yes}} \approx 0.97$ under Hard-CE (penalty 0.24 at $\beta_c=8$) but $\hat{p}_{\text{yes}} \approx 0.65$ under Soft-KL (penalty 2.8), collapsing the planner’s contrast signal across 91.7% of Pittsburgh edges. This is why 88–91% of random OD trips can be meaningfully rerouted under Soft-KL, matching calibration-aware deployment guidance for embodied agents [30].

Foundation encoders for robot perception, geometry over language. DINOv2-large leads Soft-KL while CLIP-B/32 leads Hard-CE. This ranking inversion is consistent with traversability depending on geometric cues such as slope, crack width, surface continuity, and obstacle clearance, where dense self-supervised features are more useful than language-aligned surface priors.

Hard labels fail under subjective robot-learning signal. The walking-cane result shows that Hard-CE fails because of target encoding, not encoder choice. For robot-learning tasks based on demonstrations, preferences, or crowd judgments, majority-vote labels can erase the bimodality a downstream planner needs; distribution-valued targets preserve it.

Why fine-tune a VLM rather than prompt one. Zero-shot VLMs come within 0.041 F1 of our best probe but are $6\times$ worse calibrated and $\sim 80\times$ slower. For a planner, the calibration gap is decisive: overconfident binary outputs collapse the routing gradient even when argmax accuracy is competitive. Fine-tuning with the same KL objective is the direct path to closing that gap, especially in small-data regimes [28].

8 Limitations and Responsible Deployment

Visual sample and statistical scope. Training uses 52 GSV views from ten cities (3,075 probe parameters per aid for DINOv2-large). The calibration claim rests on the Brier ratio between objectives, not absolute F1, and encoder rankings within overlapping CIs are suggestive rather than definitive.

Geographic and demographic coverage. Taipei transfer is near chance (+1.8 pp), making Western-centric data the headline limitation. Demographics beyond aid type were not collected, and aid classes conflate equipment geometry, motor power, and individual risk tolerance.

Single routing city, offline evaluation. Routing uses one Pittsburgh OSM snapshot; the ORS comparison is $n=50$ pairs and should be read as directional evidence. We have not run a live wheelchair-user study, evaluated transient obstructions or sensor noise, or tested batched/low-power deployment.

Responsible deployment. The per-aid probability is a community-level prior, not a per-user safety guarantee. A deployed system should use it only for route ranking, verify the curbside scene with onboard sensors [2], expose uncertainty rather than an argmax, retain user-specific β_c , and avoid regions where the transfer probe is near chance.

9 Conclusion

We presented a learned perception module that treats vote distributions from 829 mobility-aid users as first-class training targets and produces calibrated per-aid passability probabilities for autonomous mobility planning. A frozen-encoder probe trained by KL divergence achieves a $3.2\times$ soft-Brier improvement over hard cross-entropy across eight backbones; composed into risk-aware Dijkstra, it improves passability on 87.8–91.0% of 10,000 Pittsburgh OD pairs at 7.4–9.8% distance overhead and outperforms OpenRouteService’s wheelchair profile on every aid. The central lesson is that when subjective human judgment supervises a downstream policy, calibration—not argmax accuracy—is the operative metric. Code and pre-extracted features will be released upon acceptance.

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441 The appendices report the full per-encoder, per-aid, and per-OD-pair breakdowns that support the
 442 headline numbers in the main paper. Tables, figures, and equations are numbered with the prefix of
 443 their parent appendix section (e.g., Fig. A.1, Table B.3).

444 A Dataset and Annotator Disagreement

445 This appendix details the annotator-vote distributions that motivate our soft-label training: per-pair
 446 entropy across all five mobility aids, and representative Project Sidewalk audit views illustrating
 447 per-aid disagreement.

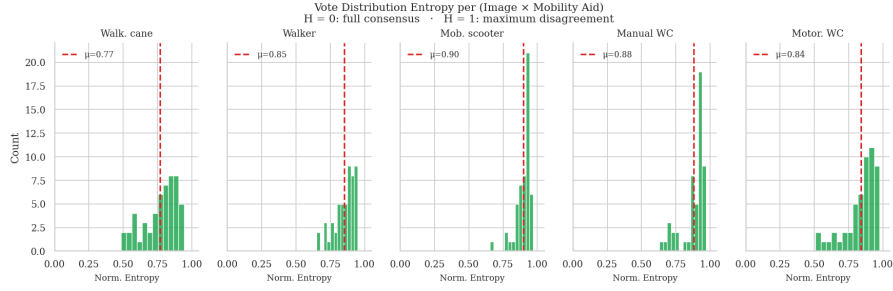
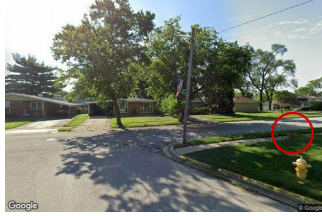
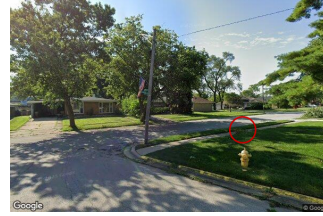


Figure A.1: Normalised vote entropy per (image, aid) pair for all five mobility aids. Dashed red line = per-aid mean. Consistently high entropy across all aids confirms that soft-label training is warranted and that hard majority labels discard genuine information.



(a) Sparse cue: missing curb ramp at intersection. Manual WC “no” = 78%.



(b) Similar scene, but the lawn-strip transition is passable. Manual WC “yes” = 61%.



(c) Clean continuous concrete; high inter-aid agreement on “yes.”



(d) Driveway obstruction. Manual WC “no” = 56%, cane “yes” = 64%.

Figure A.2: Representative Project Sidewalk audit views illustrating per-aid disagreement. Red circles mark the focal region. Adjacent panoramas with near-identical global appearance can produce sharply different per-aid vote distributions.

448 B Full Per-Encoder Per-Aid Calibration Results

449 This appendix expands Table 1 by reporting per-aid macro-F1, soft Brier, and Expected Calibration
 450 Error (ECE) for every encoder under both training objectives. All numbers are 5-fold panorama-
 451 level CV at seed = 42, identical to the main paper. Bold cells highlight the best encoder per aid
 452 within each objective. Figs. B.1 and B.2 visualise the same data.

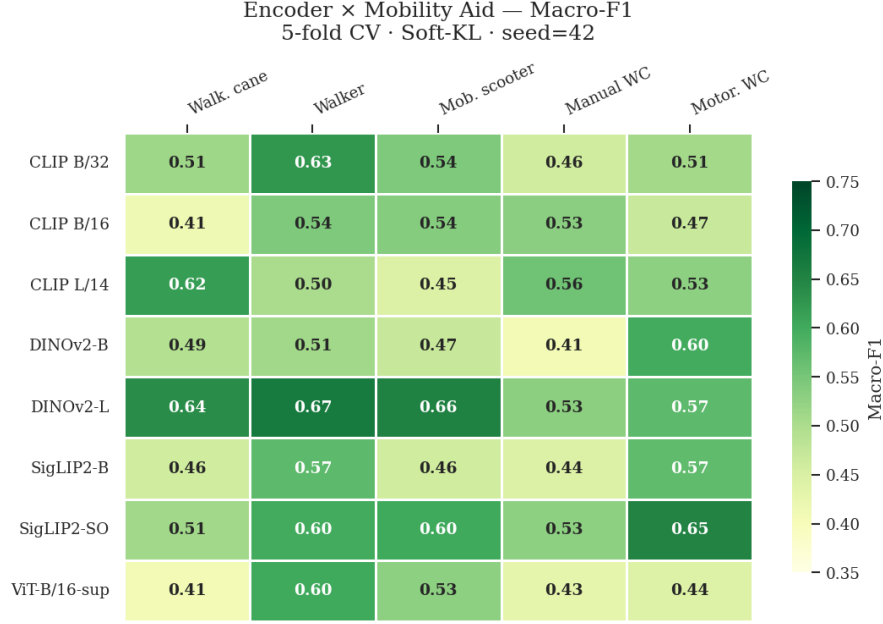


Figure B.1: Encoder × mobility aid macro-F1 heatmap under Soft-KL. DINOv2-large shows the most consistent per-aid coverage; walking cane is the hardest aid for every encoder, reflecting its 6.4:1 yes-to-no imbalance.

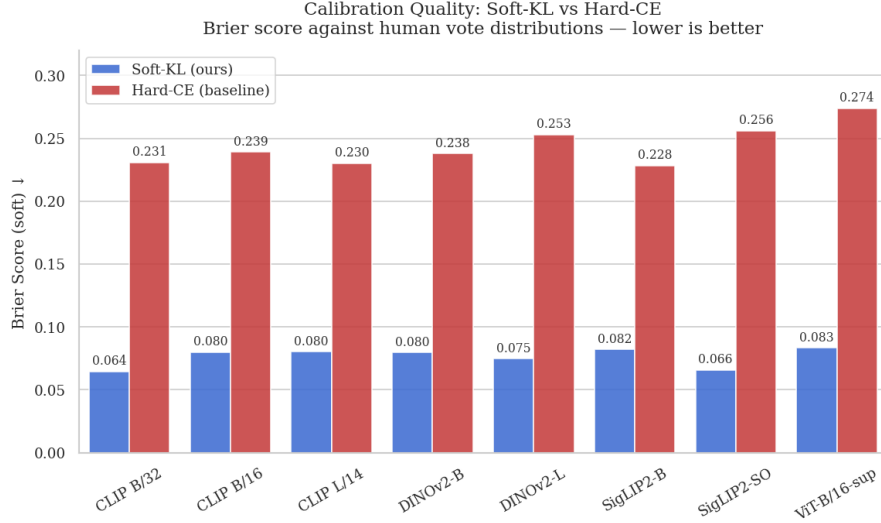


Figure B.2: Soft Brier score per encoder: Soft-KL (blue) vs. Hard-CE (red). Lower is better. The $\sim 3\times$ gap is consistent across all eight encoders; Hard-CE produces overconfident predictions that misrepresent genuine human uncertainty.

Table B.1: Soft-KL: per-aid macro-F1. Best per column in bold.

Encoder	Cane	Walker	Scooter	Man.WC	Mot.WC	Mean
CLIP B/32	0.513	0.625	0.544	0.459	0.509	0.530
CLIP B/16	0.414	0.542	0.537	0.532	0.469	0.499
CLIP L/14	0.618	0.503	0.446	0.557	0.529	0.531
DINOv2-base	0.493	0.510	0.470	0.408	0.599	0.496
DINOv2-large	0.640	0.667	0.656	0.531	0.573	0.613
SigLIP2-base	0.461	0.574	0.459	0.444	0.571	0.502
SigLIP2-SO400M	0.511	0.600	0.601	0.530	0.652	0.579
ViT-B/16-sup	0.408	0.603	0.529	0.429	0.442	0.482

Table B.2: Hard-CE: per-aid macro-F1. Best per column in bold.

Encoder	Cane	Walker	Scooter	Man.WC	Mot.WC	Mean
CLIP B/32	0.453	0.761	0.558	0.675	0.639	0.617
CLIP B/16	0.434	0.536	0.569	0.562	0.618	0.544
CLIP L/14	0.453	0.707	0.698	0.523	0.597	0.595
DINOv2-base	0.446	0.720	0.668	0.574	0.664	0.614
DINOv2-large	0.489	0.675	0.746	0.502	0.652	0.613
SigLIP2-base	0.458	0.700	0.587	0.572	0.664	0.596
SigLIP2-SO400M	0.453	0.669	0.519	0.483	0.652	0.555
ViT-B/16-sup	0.513	0.682	0.606	0.553	0.509	0.573

Table B.3: Soft-KL: per-aid soft Brier score ($\downarrow$). Lower is better (perfect calibration = 0). All values < 0.15 , vs. Hard-CE values ≥ 0.19 in Table B.4.

Encoder	Cane	Walker	Scooter	Man.WC	Mot.WC	Mean
CLIP B/32	0.051	0.066	0.044	0.059	0.103	0.064
CLIP B/16	0.059	0.085	0.048	0.070	0.139	0.080
CLIP L/14	0.061	0.076	0.068	0.087	0.109	0.080
DINOv2-base	0.072	0.079	0.063	0.082	0.102	0.080
DINOv2-large	0.061	0.070	0.051	0.086	0.106	0.075
SigLIP2-base	0.079	0.082	0.057	0.089	0.104	0.082
SigLIP2-SO400M	0.062	0.069	0.047	0.061	0.090	0.066
ViT-B/16-sup	0.065	0.082	0.061	0.075	0.133	0.083

Table B.4: Hard-CE: per-aid soft Brier score ($\downarrow$). Uniformly 3 to $4\times$ worse than Soft-KL across all (encoder, aid) cells.

Encoder	Cane	Walker	Scooter	Man.WC	Mot.WC	Mean
CLIP B/32	0.229	0.213	0.234	0.243	0.234	0.231
CLIP B/16	0.212	0.253	0.205	0.245	0.280	0.239
CLIP L/14	0.200	0.221	0.239	0.249	0.240	0.230
DINOv2-base	0.195	0.208	0.265	0.291	0.230	0.238
DINOv2-large	0.211	0.235	0.242	0.304	0.272	0.253
SigLIP2-base	0.197	0.231	0.245	0.235	0.234	0.228
SigLIP2-SO400M	0.208	0.265	0.244	0.270	0.293	0.256
ViT-B/16-sup	0.218	0.288	0.268	0.259	0.336	0.274

Table B.5: Expected Calibration Error (ECE, $\downarrow$). Soft-KL achieves lower ECE than Hard-CE on the same encoder in 7 of 8 cases. Despite both having moderate ECE, Brier (which penalises probability magnitude, not just rank) is the more discriminative metric (cf. Tables B.3 and B.4).

Encoder	Walking cane		Manual WC	
	Soft-KL	Hard-CE	Soft-KL	Hard-CE
CLIP B/32	0.270	0.179	0.202	0.238
CLIP B/16	0.275	0.205	0.214	0.219
CLIP L/14	0.223	0.166	0.258	0.269
DINOv2-base	0.234	0.165	0.210	0.250
DINOv2-large	0.243	0.185	0.274	0.303
SigLIP2-base	0.193	0.192	0.262	0.225
SigLIP2-SO400M	0.277	0.183	0.239	0.272
ViT-B/16-sup	0.209	0.194	0.262	0.291

C Zero-Shot VLM Analysis

This appendix expands the zero-shot VLM bottom block of Table 1 with the inference-latency Pareto frontier (Fig. C.2), an aggregated F1 comparison (Fig. C.1), and full per-prompt sensitivity across three VLMs (Table C.1).

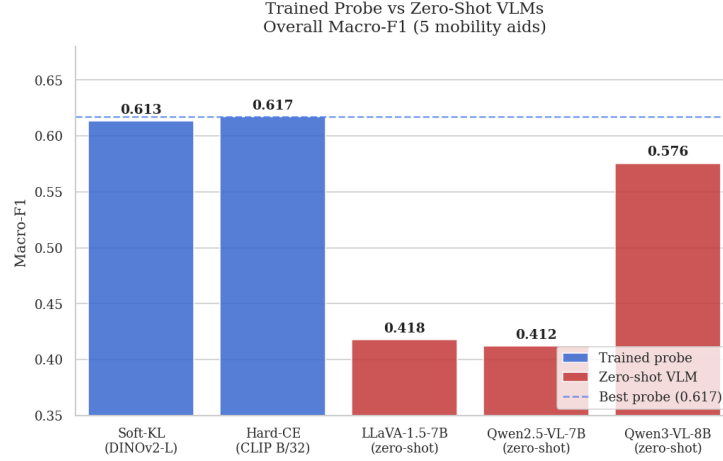


Figure C.1: Trained probes vs. zero-shot VLMs on overall macro-F1. Dashed blue line = best probe (F1 = 0.617). Qwen3-VL-8B (F1 = 0.576) comes within 0.041 F1 at $\sim 80\times$ higher latency and $6.2\times$ worse Brier.

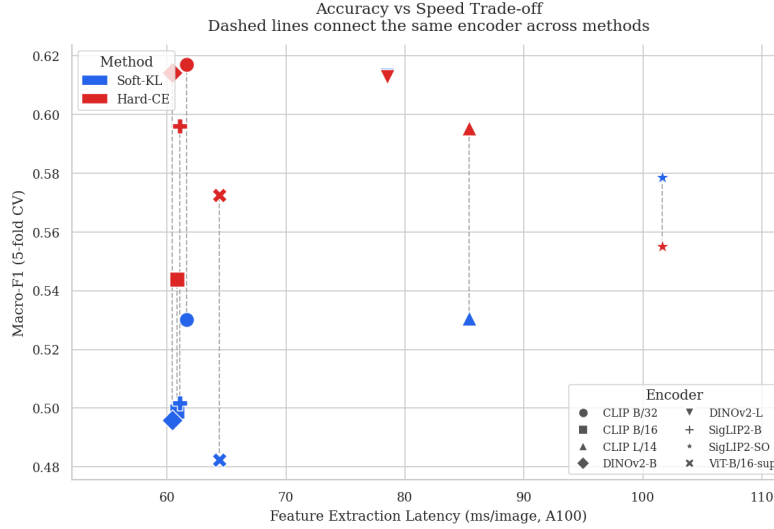


Figure C.2: Macro-F1 vs. inference latency for all 8 encoders under Soft-KL (blue) and Hard-CE (red). DINOv2-large is Pareto-optimal under Soft-KL (78.5 ms, F1 = 0.613). All encoder probes fit a real-time routing budget.

Table C.1: Full prompt-sensitivity matrix. Three prompts (Direct / Descriptive / CoT) $\times$ three VLMs. Best per model in bold. Parse failures = number of responses (out of 260) lacking a parseable ternary label. Only CoT on Qwen2.5-VL-7B incurred any parse failures (2/260), confirming that prompt-engineering effects do not reduce to format-compliance differences.

Model	Variant	F1	Bal.Acc	Parse fail
Qwen3-VL-8B	Direct	0.513	0.532	0
	Descriptive	0.576	0.633	0
	CoT	0.547	0.563	0
LLaVA-1.5-7B	Direct	0.488	0.520	0
	Descriptive	0.430	0.514	0
	CoT	0.542	0.563	0
Qwen2.5-VL-7B	Direct	0.442	0.610	0
	Descriptive	0.412	0.596	0
	CoT	0.479	0.622	2

457 D Walking-Cane Failure Cases

458 This appendix supports the walking-cane failure analysis in §5. Figure D.1 shows the full vote distri-
 459 butions for the seven majority-“no” walking-cane images that drive the Hard-CE collapse described
 460 in the main paper.

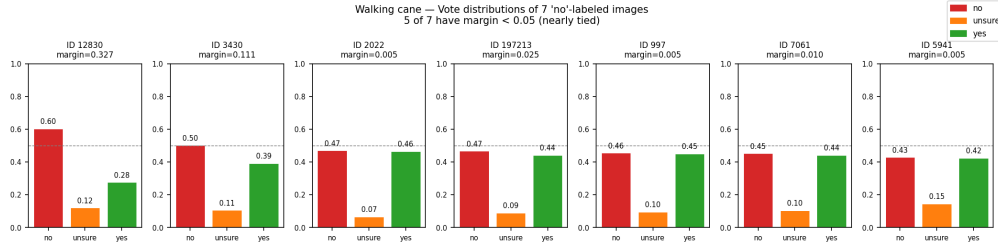


Figure D.1: The seven majority-“no” walking-cane images and their vote distributions. IDs 2022, 197213, 997, 7061, and 5941 all have margin < 0.05 between the top two classes (“no” and “yes”). Hard-CE training treats these as confident negatives and conflicts with the 45 confident positives in the rest of the cane distribution. Soft-KL retains the empirical bimodality and predicts $[\hat{p}_{\text{no}} \approx 0.4, \hat{p}_{\text{unsure}} \approx 0.1, \hat{p}_{\text{yes}} \approx 0.5]$, the calibrated correct answer that explains the +42% relative F1 gain.

461 E Geographic Transfer: Per-City Detail

462 This appendix expands the Geographic transfer probe paragraph in §5. Figure E.1 shows the per-city
 463 balanced accuracy across Pittsburgh, Detroit, Zurich, and Taipei; Table E.1 reports the underlying
 464 counts, baselines, and gains.

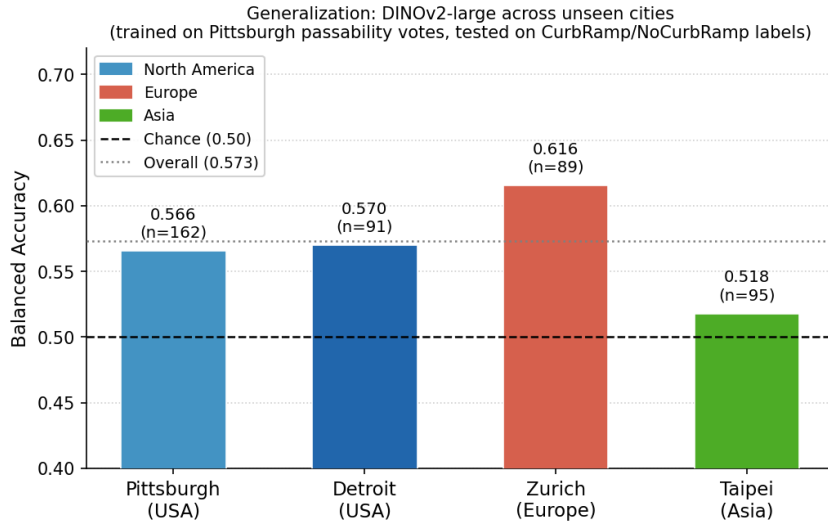


Figure E.1: Per-city balanced accuracy for DINOv2-large + Soft-KL on binary CurbRamp labels in unseen cities. Dashed line = chance (0.50). Pittsburgh/Detroit consistency rules out city-specific overfitting; Zurich gains +11.6 pp (Amsterdam in training); Taipei is only +1.8 pp above chance, the Western-centric-data limit.

Table E.1: Per-city generalization details. DINOv2-large + Soft-KL trained on the 52-view passability benchmark, evaluated on Project Sidewalk binary CurbRamp labels in each city. “Maj.” = always-predict-majority-class baseline.

City	Region	n	Bal.Acc	Maj.	Gain (pp)	Notes
Pittsburgh	N.America	162	0.566	0.500	+6.6	in-distribution
Detroit	N.America	91	0.570	0.500	+7.0	similar Midwest US
Zurich	Europe	89	0.616	0.500	+11.6	strong transfer
Taipei	Asia	95	0.518	0.500	+1.8	near-chance
All	n/a	437	0.573	0.500	+7.3	weighted mean

F Routing: Pareto Analysis

This appendix supports two routing claims in §6: (i) the choice $\beta_c=8$ and (ii) the Monte Carlo distributional behaviour. Table F.1 and Fig. F.1 give the per-aid Pareto sweep over $\beta_c \in \{1, 2, 4, 8, 16\}$; the inflection between $\beta_c = 4$ and $\beta_c = 8$ produces the largest accessibility gain per unit of distance overhead. Table F.2 and Figs. F.2, F.3 report the full P25/P50/P75 distribution of the 10,000-pair Monte Carlo evaluation; distance overhead is right-skewed (median $< 1\%$ for all aids, while the mean is 4–5%).

Table F.1: Barrier-cost sweep on the canonical OD pair. Each cell shows $(\Delta\bar{p}_{yes}, \Delta\ell \%)$. Bold marks the chosen operating point.

β_c	Cane	Walker	Scooter	Man.WC	Mot.WC
1	(0.029, 0.6)	(0.000, 0.0)	(0.000, 0.0)	(0.000, 0.0)	(0.000, 0.0)
2	(0.029, 0.6)	(0.026, 0.6)	(0.000, 0.0)	(0.000, 0.0)	(0.000, 0.0)
4	(0.145, 17.8)	(0.144, 24.8)	(0.140, 17.8)	(0.130, 17.8)	(0.133, 17.8)
8	(0.155, 24.8)	(0.144, 24.8)	(0.140, 17.8)	(0.134, 19.3)	(0.146, 24.8)
16	(0.155, 24.8)	(0.144, 24.8)	(0.146, 24.8)	(0.140, 24.8)	(0.146, 24.8)

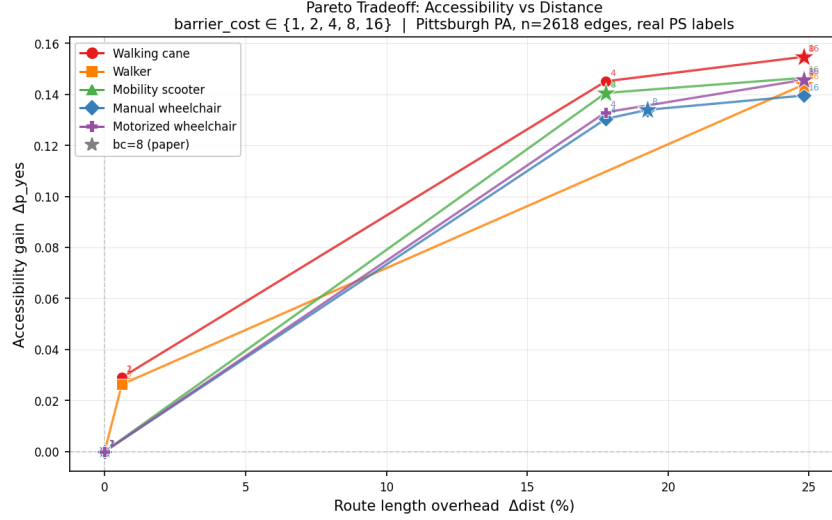


Figure F.1: Pareto curve for $\beta_c \in \{1, 2, 4, 8, 16\}$, averaged across the five mobility aids. The accessibility gain saturates past $\beta_c = 8$, justifying our operating point.

Table F.2: Full Monte Carlo distribution (10,000 OD pairs). P25, P50, P75 of $\Delta\bar{p}_{yes}$ and $\Delta\ell\%$, plus the fraction of pairs improved. Pittsburgh OSM graph, $\beta_c = 8$, seed = 42, min separation 200 m.

Aid	$\Delta\bar{p}_{yes}$			$\Delta\ell \%$			% impr.	$\bar{p}_{yes}^{std}$	$\bar{p}_{yes}^{acc}$
	P25	P50	P75	P25	P50	P75			
Walking cane	0.000	0.021	0.072	0.0	0.77	n/a	60.6	0.705	0.748
Walker	0.000	0.020	0.085	0.0	0.39	n/a	59.2	0.625	0.673
Mobility scooter	0.000	0.027	0.094	0.0	0.48	n/a	60.0	0.570	0.623
Manual wheelchair	0.000	0.030	0.104	0.0	0.69	n/a	60.5	0.499	0.558
Motorized wheelchair	0.000	0.028	0.102	0.0	0.85	n/a	61.5	0.529	0.587

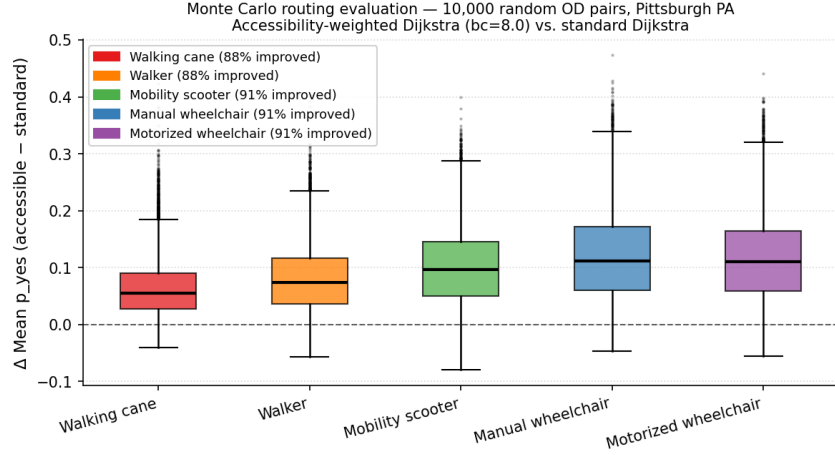


Figure F.2: Boxplot of per-aid $\Delta \bar{p}_{\text{yes}}$ on 10,000 random OD pairs. All five aids show positive median improvement; the IQR confirms the rerouting benefit is widespread, not driven by outliers.

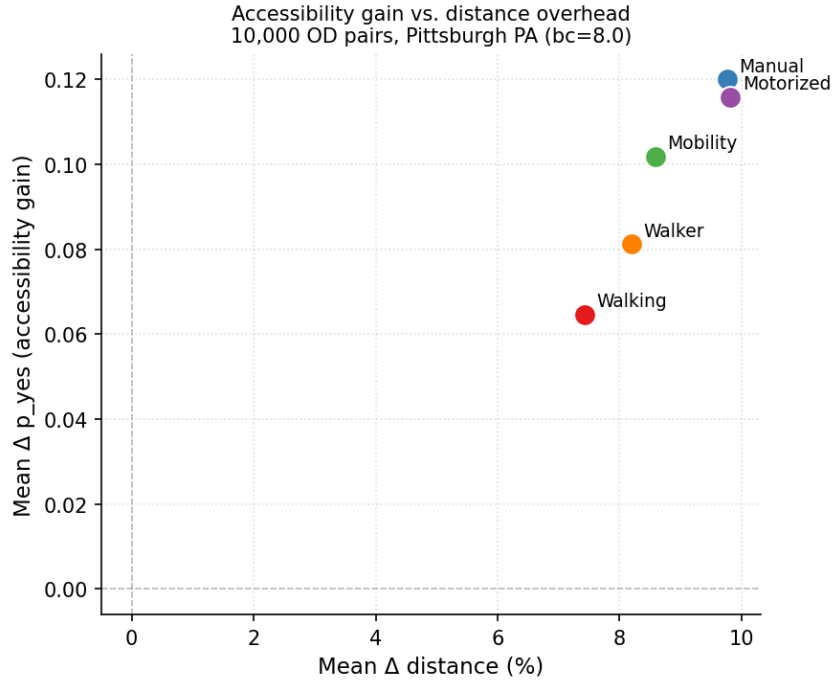


Figure F.3: Pareto scatter: mean accessibility gain $\Delta \bar{p}_{\text{yes}}$ vs. mean distance overhead across 10,000 random OD pairs. All five aids achieve positive accessibility gain at 7–10% extra distance, confirming the routing benefit extends well beyond curated examples.

472 G Routing: Per-Aid Visualisations and Comparisons

473 This appendix supports the routing claims in §6 with three pieces of evidence: (i) per-aid optimal
474 paths for the canonical OD pair (Fig. G.1, Fig. G.2); (ii) the full Std/ORS/Ours per-aid breakdown
475 for the 40 routable Pittsburgh pairs (Table G.1); and (iii) the Hard-CE vs Soft-KL routing cross-
476 evaluation on the canonical OD pair (Table G.2, Fig. G.3).

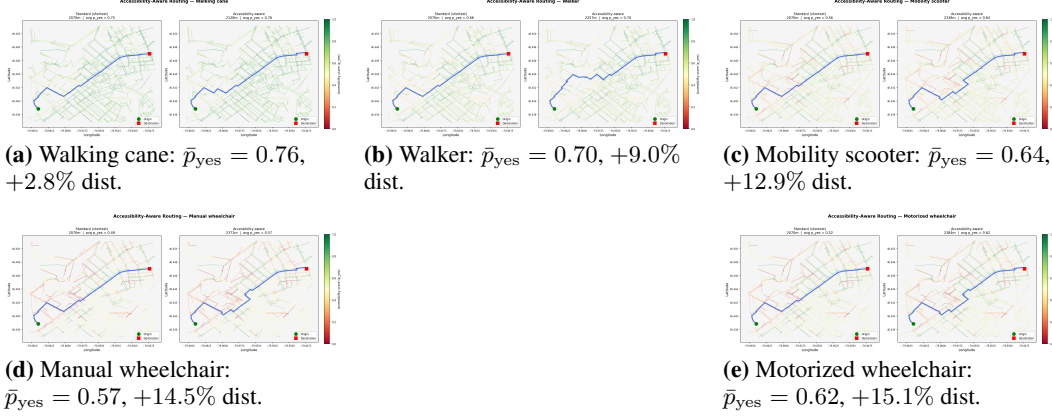


Figure G.1: Per-aid accessible routes (canonical OD pair, $\beta_c = 8$). The five panels reveal that the optimal accessible path is aid-specific: walking-cane and walker (top, easier aids) take the most efficient detour; manual and motorized wheelchairs (bottom) prefer further-removed but smoother corridors.

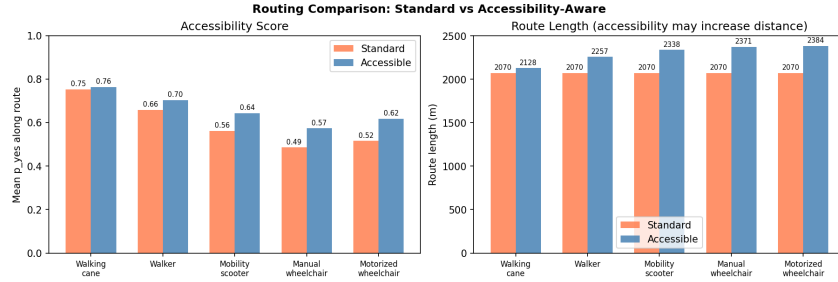


Figure G.2: Per-aid summary for the canonical OD pair: accessibility score (left) and total route length (right). Walking cane (already 0.75 baseline) gains the least in absolute $\hat{p}_{\text{yes}}$ but the most in routing stability; the wheelchair aids gain +0.07 to +0.13 at +12 to +18% distance.

Table G.1: ORS wheelchair-profile comparison (full breakdown). 40 routable OD pairs out of 50. “ORS–Std.” = gain ORS achieves over plain Dijkstra; “Ours–ORS” = our additional gain on top of ORS. Our method yields higher mean $\hat{p}_{\text{yes}}$ than ORS on every aid in this Pittsburgh snapshot by 0.034 to 0.047. We queried the public OpenRouteService API /v2/directions/wheelchair endpoint between 2026-05-12 and 2026-05-14; the 10 unroutable pairs returned HTTP 400 with “Could not find routable point within radius.”

Aid	Std.	ORS	Ours	ORS–Std.	Ours–ORS
Walking cane	0.704	0.712	0.748	+0.009	+0.036
Walker	0.623	0.634	0.668	+0.011	+0.034
Mobility scooter	0.573	0.585	0.625	+0.012	+0.040
Manual wheelchair	0.505	0.519	0.566	+0.013	+0.047
Motorized wheelchair	0.532	0.545	0.590	+0.013	+0.045
Mean	0.587	0.599	0.639	+0.012	+0.040

Table G.2: Hard-CE vs Soft-KL routing cross-evaluation on the canonical OD pair. All $\hat{p}_{\text{yes}}$ values evaluated by the calibrated Soft-KL metric (our reference risk estimate), except the last column which shows Hard-CE’s own (inflated) self-assessment. Gap = Soft-KL route minus Hard-CE route by the calibrated metric. Hard-CE routes are 430–436 m vs. 412–575 m for Soft-KL; despite being shorter, they are consistently less accessible.

Aid	Std (calibrated)	SKL route	HCE route	Gap (SKL–HCE)	HCE self-report	HCE overestimate
Walking cane	0.516	0.671	0.657	+0.014	0.721	+0.064
Walker	0.486	0.630	0.600	+0.030	0.684	+0.085
Mobility scooter	0.459	0.600	0.587	+0.013	0.675	+0.089
Manual wheelchair	0.442	0.576	0.561	+0.015	0.660	+0.099
Motorized wheelchair	0.446	0.592	0.572	+0.020	0.625	+0.053

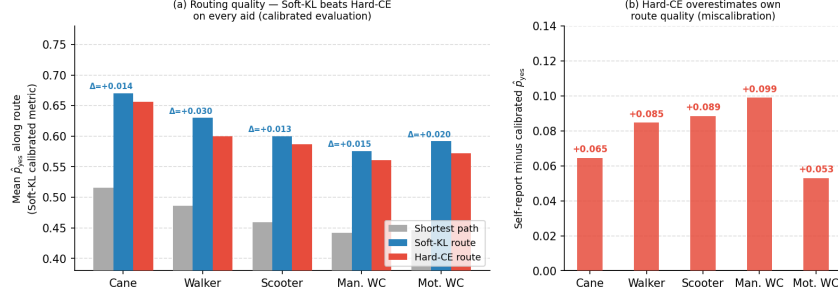


Figure G.3: Cross-evaluation of routing under Soft-KL vs. Hard-CE edge scores on the canonical Pittsburgh OD pair. Bars show mean $\hat{p}_{yes}$ along each route, evaluated by the calibrated Soft-KL metric. Hard-CE routes (solid red) are uniformly lower-scoring than Soft-KL routes (blue) on the calibrated metric, despite Hard-CE’s own inflated self-report (hatched red).

477 H Algorithms and Reproducibility

478 This appendix gives pseudocode for the two algorithms underlying the paper and the com-
479 pute/reproducibility details required to replicate every number in the main text.

480 H.1 Soft-KL Probe Training

Algorithm 1 Soft-KL training of a per-aid linear probe

Require: Frozen encoder E , training images $\{x_i\}_{i=1}^N$, vote distributions $\{y_{i,a}\}$ for $a \in \text{Aids}$

- 1: Pre-compute $\mathbf{f}_i = E(x_i)/\|E(x_i)\|_2$ once per image
 - 2: Initialise W_a, b_a from $\mathcal{N}(0, 0.01)$ for each aid a
 - 3: **for** epoch = 1 to 50 **do**
 - 4: **for** mini-batch $\mathcal{B} \subset \{1, \dots, N\}$ of size 64 **do**
 - 5: $\hat{y}_{i,a} = \text{softmax}(W_a \mathbf{f}_i + b_a)$
 - 6: $w_{i,a} = 1 - H(y_{i,a})/H_{\max}$
 - 7: $\mathcal{L} = \sum_{i \in \mathcal{B}} \sum_a w_{i,a} D_{\text{KL}}(y_{i,a} \parallel \hat{y}_{i,a})$
 - 8: Update $\{W_a, b_a\}$ via AdamW (lr 10^{-3} , wd 10^{-4})
 - 9: **end for**
 - 10: **end for**
 - 11: **return** $\{W_a, b_a\}_{a \in \text{Aids}}$
-

481 H.2 Accessibility-Aware Dijkstra

Algorithm 2 Per-aid accessibility-aware shortest path

Require: Pedestrian graph $G = (V, E)$, edge lengths ℓ_e , per-aid per-edge probabilities $\hat{p}_{yes,e}^a$, origin s , destination t , aid a , barrier cost β_c

- 1: **for** $e \in E$ **do**
 - 2: $w_e^a \leftarrow \ell_e \cdot (1 + \beta_c (1 - \hat{p}_{yes,e}^a))$
 - 3: **end for**
 - 4: $P \leftarrow \text{Dijkstra}(G, s, t, w^a)$
 - 5: $\bar{p}_{yes}^a(P) \leftarrow \frac{\sum_{e \in P} \ell_e \hat{p}_{yes,e}^a}{\sum_{e \in P} \ell_e}$
 - 6: **return** $P, \bar{p}_{yes}^a(P), \sum_{e \in P} \ell_e$
-

482 H.3 Compute and Reproducibility

483 **Hardware.** All experiments were run on a single NVIDIA A100 40 GB GPU. Encoder feature
484 extraction for all 200 verified image crops takes 4 to 10 s per encoder (DINOv2-large is the slowest
485 at 9.8 s). Soft-KL or Hard-CE probe training for the full 5-fold CV across one encoder takes 30 to
486 90 s. The full 8-encoder $\times$ 2-loss CV grid runs end-to-end in < 30 minutes.

487 **Software.** Python 3.9, PyTorch 2.2, transformers 4.40, timm 0.9.16, NetworkX 3.2, OSMnx 1.9.
488 The OpenRouteService queries used the public API at [https://api.openrouteservice.org/](https://api.openrouteservice.org/v2/directions/wheelchair)
489 [v2/directions/wheelchair](https://api.openrouteservice.org/v2/directions/wheelchair) on 2026-05-12 through 2026-05-14. Polyline decoding uses the
490 polyline pip package v2.0.

491 **Random seeds.** All experiments use the same global seed (42) for NumPy, PyTorch (CPU +
492 CUDA), Python random, and the panorama-level fold split. The Monte Carlo OD-pair sampler
493 reseeds NumPy with the same seed before generating its 10,000 pairs.

494 **Code availability.** Code and pre-extracted features will be released on acceptance under an MIT
495 licence, including: (i) the per-encoder feature extraction pipeline; (ii) Soft-KL / Hard-CE probe
496 training; (iii) the accessibility-aware Dijkstra implementation; (iv) the ORS comparison harness; (v)
497 the Monte Carlo evaluation; and (vi) all figure-generation notebooks.